

MIHIR PRAJAPATI

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EDUCATION

New York University (NYU), Courant Institute of Mathematical Sciences

September 2023 - May 2025

Master of Science in Computer Science

New York, USA

Relevant Courses: Big Data & Predictive Analytics, Advanced Databases, Operating Systems, Cloud & Machine Learning (ML), DevOps

Nirma University, Institute of Technology

July 2019 - May 2023

Bachelor of Technology in Computer Science and Engineering

Ahmedabad, India

Relevant Courses: Object-Oriented Programming, Computer Networks, Data Science, Information Retrieval, Computer Architecture

SKILLS

Languages: C++, Java, Python, Typescript, Javascript, C#, HTML/CSS, Bash scripting, Rust, Golang, Scala, Sql

Tools: Jenkins, Terraform, Git, Docker, Kubernetes, Ansible, Jira, Linux, Prometheus, Postman, jQuery, Red Hat OpenShift, Grafana

Frameworks: SpringBoot, CUDA, OpenMP, MPI, GraphQL, TensorFlow, PyTorch, Keras, React, Angular, Node.js, Django, Flask, Rest API, Ruby on Rails, Vue.js, Next.js

Data Technologies: MySQL, PostgreSQL, MongoDB, Oracle, AWS (DynamoDB, RDS, Redshift) Cassandra, Kafka, ELK, Redis

Cloud Platforms: AWS (EKS, S3, IAM, EC2, SQS, ELB, ECS, CloudFormation), Azure (Blob Storage, ARMTemplate)

EXPERIENCE

Ajmera Infotech

May 2024 - September 2024

Software Developer

Ahmedabad, India

- Led the design and deployment of a scalable distributed caching layer using **Redis**, **Kubernetes**, and **Spring Boot**, improving API throughput by 15% and enabling smoother CI/CD-based rollouts across containerized environments.
- Developed and maintained a **Cucumber-based BDD** test framework, integrating **Karate**, **Selenium**, and **JUnit** to validate UI performance, memory stability, and overall service health under high concurrency.
- Built resilient data pipelines connecting **Cassandra**, **MySQL**, and **S3** using **Spring Batch** and multithreaded ETL tasks, streamlining reporting pipelines for internal **GraphJS**-based dashboards.

NikSan Tech

August 2022 - August 2023

Software Engineer

Ahmedabad, India

- Built scalable microservices and full-stack apps using **Spring Boot**, **Java 8**, **Angular**, **Hibernate**, **JWT**, and MFA-secured REST APIs; improved scalability across 5+ products and reduced **MTTR** by 30% with CI/CD via **Jenkins**, **Docker**, and **Kubernetes**.
- Developed **Kafka**-based async services and **Spark** ETL pipelines processing 500K+ daily records with Parquet/Hive support; optimized **MongoDB** queries to reduce **data retrieval** time by 70%.
- Automated testing with **Cypress**, **Bash**, and **Python**; improved reliability with **JUnit**, **Mockito**, **Splunk**, and **AWS CloudWatch**, cutting manual QA by 50% and boosting performance by 17%.

Ajmera Infotech

June 2022 - August 2022

Software Developer Intern

Ahmedabad, India

- Developed and deployed a chatbot using **Python**, **Spacy**, **React Native**, **Kotlin**, and **Docker**, improving user engagement by 62% and creating comprehensive API documentation using **Swagger** to facilitate seamless integration.
- Tested and ensured cross-platform functionality for mobile and web apps, leveraging **Node.js** and **Express.js** for web functionality, **Postman** and **JUnit** for performance and scalability testing, and **AWS Lambda** for deployment in a serverless architecture.

PROJECTS

E-Commerce Microservice

- Built a **React** and **Flask**-based microservice, implementing **RESTful APIs** and utilizing **SQLAlchemy** with **PostgreSQL** for database.
- Streamlined **CI/CD** with **GitHub Actions**, automating **Kubernetes** deployments on **OpenShift**, cutting deployment time by 50%.
- Ensured code quality with **Pytest**, **BDD** tests via **Selenium**, and improved performance with **Redis** caching.

Retrieval-Augmented Generation (RAG) Recommender

- Developed a research assistant using **LangGraph**, **LangChain**, **OpenAI embeddings**, and **ChromaDB** for semantic paper retrieval.
- Integrated **Model Context Protocol (MCP)** to manage tool calls and structured agent reasoning.
- Tracked and refined LLM workflows using **LangSmith**, improving response quality and interpretability.
- Deployed a dockerized **Flask app** on **Kubernetes** with tuning via **Kubeflow**. [\[Link\]](#)

Replicated Concurrency and Recovery System

- Implemented a **distributed database** with **snapshot isolation**, focused on replication and **failure recovery**. Simulated a complex environment with 20 variables across 10 sites, ensuring integrity and robustness against failures.

Kernext

- Built a Linux **Kernel** in **C++**, implementing **process** and **memory management** via **Linux system calls**, supporting **50+ concurrent processes**.
- Created a **priority-based, multilevel scheduler**, reducing **context switching** by **30%** and improving **CPU utilization**.
- Optimized **virtual memory** with **paging**, **segmentation**, and **demand paging**, boosting **allocation efficiency** by **25%**.